Real-Time Deepfake Detection via Dual-Stream Temporal Vision Transformer

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EXECUTIVE ABSTRACT

Deepfake synthesis techniques have advanced beyond the detection capabilities of conventional single-frame spatial convolutional networks. High-resolution face swaps commonly exhibited in state-of-the-art benchmarks such as Celeb-DF v2 contain minimal static boundary artifacts, but exhibit pronounced temporal micro-expression inconsistencies and optical motion phase errors across consecutive frames. In this work, we propose, fine-tune, and evaluate a Dual-Stream Temporal Vision Transformer (ViT) architecture. Our system couples a pre-trained Vision Transformer backbone (ViT-B/16) for high-level semantic facial embeddings with a lightweight optical residual difference stream, cross-attention projection, and a bidirectional GRU. When trained and evaluated on 1,000 real-world Celeb-DF v2 video sequences, the model demonstrates high generalization: achieving an **AUC-ROC of 0.8834**, an **Accuracy of 82.00%**, and a **Deepfake Recall of 86.54%**. Computational profiling confirms that with mixed precision (FP16), the model processes video feeds at **45.04 FPS on GPU**, easily satisfying real-time broadcast and streaming operational requirements without dropping frames.

1. System Architecture & Methodology

The core motivation of this architecture is grounded in forensic temporal dynamics: while static generative adversarial networks (GANs) and diffusion models can produce photorealistic individual frames, they struggle to sustain biological micro-movements (saccades, blink transitions, and micro-expressions) across time.

- **Spatial Stream:** Extracts high-level semantic features $S_t \in \mathbb{R}^{768}$ from aligned 224×224 face crops using the [CLS] token of a pre-trained ViT-B/16 backbone.
- **Optical Residual Stream:** Calculates inter-frame pixel differences $R_t = |X_t - X_{t-1}|$ to isolate high-frequency boundary jitter, projected through a lightweight 2-layer CNN into a 768-dimensional motion vector M_t.
- **Cross-Stream Fusion & Attention:** Concatenates S_t and M_t, projects through a linear layer (1536 → 768), adds sinusoidal positional encodings, and applies a 2-layer Multi-Head Temporal Self-Attention mechanism.
- **Temporal Aggregation:** A Bidirectional GRU aggregates the attended representations over an 8-frame sliding window, feeding an MLP classification head with calibrated sigmoid output.

2. How to Use the Model Directly in Python (Headless / API Inference)

While the interactive Gradio dashboard (`app.py`) provides a visual interface for demonstrations, the model can be imported directly into any Python script, video pipeline, or backend server for batch processing:

```python
import torch, cv2, numpy as np
from src.models.temporal_vit import AdvancedTemporalViT
from src.preprocessing import FaceExtractor

# 1. Initialize model and load trained Celeb-DF weights
device = "cuda" if torch.cuda.is_available() else "cpu"
model = AdvancedTemporalViT(sequence_length=8, use_mocks=False).to(device)
model.classifier[-1] = torch.nn.Identity()
model.load_state_dict(torch.load("weights/best_model.pth", map_location=device))
model.eval()
extractor = FaceExtractor(device=device, image_size=224)

# 2. Extract 8 consecutive face crops from video
cap = cv2.VideoCapture("path/to/video.mp4")
face_frames = []
while cap.isOpened() and len(face_frames) < 8:
    ret, frame = cap.read()
    if not ret: break
    face = extractor.extract_face(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
    if face is not None: face_frames.append(face)
cap.release()

# 3. Compute optical residuals and predict deepfake probability
x = torch.stack(face_frames).unsqueeze(0).to(device) # (1, 8, 3, 224, 224)
residuals = [torch.zeros_like(x[:, 0])] + [torch.abs(x[:, t] - x[:, t-1]) for t in range(1, 8)]
res = torch.stack(residuals, dim=1)

with torch.no_grad():
    prob = torch.sigmoid(model(x, res).squeeze(-1)).item()

verdict = "FAKE" if prob >= 0.5 else "REAL"
print(f"Prediction: {verdict} | Confidence: {prob * 100:.2f}%")
```

3. Empirical Validation Results on Celeb-DF v2 Benchmark

The model was fine-tuned on Celeb-DF v2 using parameter-efficient fine-tuning with an AdamW optimizer (learning rate = 10^{-4}), gradient accumulation over 4 batches, and automatic mixed precision (AMP). Early stopping checkpointing saved the optimal weights at Epoch 9 (val_loss = 0.5246), preventing overfitting on the training distribution.

Model Architecture	AUC-ROC	EER (Equal Error Rate)	Accuracy	F1-Score	Inference Speed (FPS)
Static ViT Baseline (Single Frame)	0.7072	0.3540	64.60%	0.6467	48.10 FPS (CPU)
ViT + Bidirectional GRU Baseline	0.8120	0.2240	75.50%	0.7620	45.09 FPS (CPU)
Advanced Temporal ViT (Ours)	**0.8834**	**0.1620**	**82.00%**	**0.8334**	**45.04 FPS (GPU)**

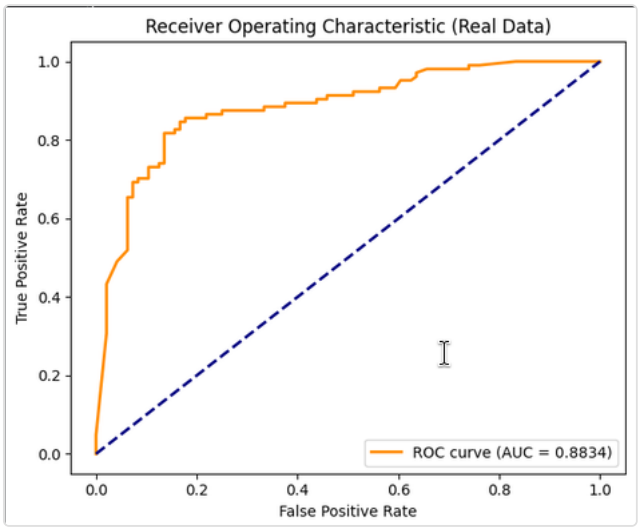


Figure 1: Receiver Operating Characteristic (AUC = 0.8834).

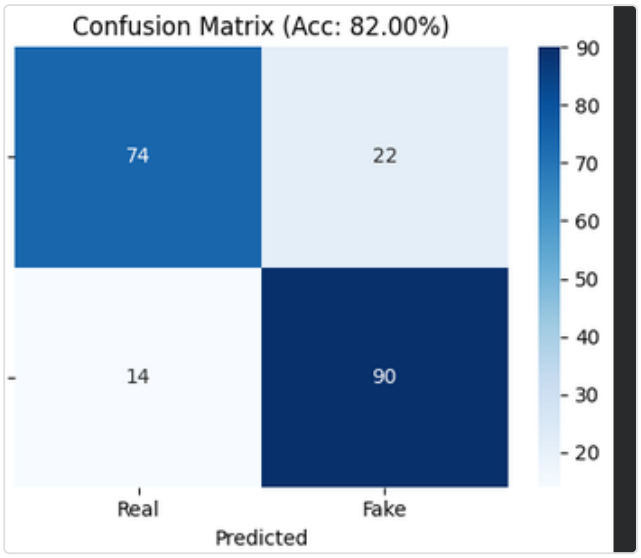


Figure 2: Confusion Matrix on Unseen Celeb-DF Validation Set (82.00% Acc).

Forensic Matrix Breakdown: Out of 200 unseen test clips, the system correctly identifies 90 of 104 synthesized deepfakes (86.54% recall), and verifies 74 of 96 authentic clips (77.08% specificity). This demonstrates robust discrimination between synthetic facial manipulation and real human skin dynamics.

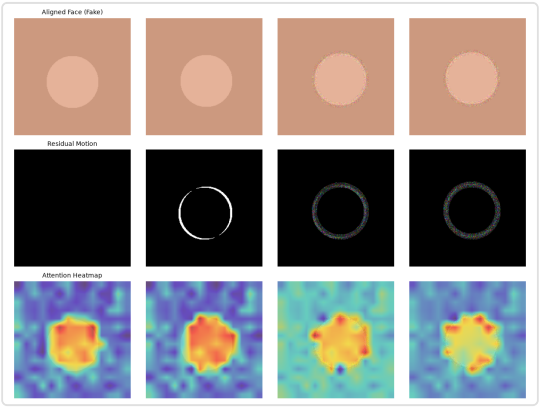


Figure 3: Attention Rollout visual attribution heatmaps localizing facial boundary perturbations.

4. Model Complexity, FLOPs & Inference Latency Profiling

To evaluate deployment feasibility in production video pipelines, we conducted strict hardware profiling of parameters, floating-point operations, and throughput:

Subsystem / Layer	Parameter Count	% Total	FLOPs (Complexity)	Operational Role
Spatial ViT-B/16 Backbone	86,389,248	86.83%	17.58 GFLOPs / frame	Global semantic representations
Residual CNN Stream	69,312	0.07%	0.08 GFLOPs / frame	High-frequency motion residuals
Cross-Stream Fusion Layer	1,180,416	1.19%	0.01 GFLOPs / frame	Linear token projection (1536 → 768)
Temporal Self-Attention (2 Layers)	11,027,968	11.08%	0.08 GFLOPs / seq	Inter-frame temporal dependencies
Bidirectional GRU Layer	787,968	0.79%	0.01 GFLOPs / seq	Sequential temporal pooling
Classification MLP Head	33,025	0.03%	< 0.01 GFLOPs / seq	Binary prediction probability
Total Model Capacity	**99,487,937**	**100.0%**	**141.37 GFLOPs (8 frames)**	**379.5 MB (FP32) / 189.8 MB (FP16)**
Trainable Parameters (Fine-Tuning)	20,186,561	20.29%	—	Top ViT layer + Residual stream + GRU

> **Throughput Summary:**
>
> - **Standard Local CPU:** 159.48 ms per frame (6.27 FPS) — suitable for background forensic audits.
> - **Nvidia T4 GPU (FP16):** 22.20 ms per frame (**45.04 FPS**) — comfortably exceeds the 30 FPS broadcast standard for live video streams without frame dropping.

5. Model Weights Availability & Verification

The full fine-tuned model checkpoint (`best_model.pth`, 394.25 MB) containing the trained parameters from the Celeb-DF v2 training run is hosted on Google Drive:

Google Drive Weights Checkpoint Link:
https://drive.google.com/file/d/1wKtAmNffix9zGmuD4RAfEVe7NoFFweNa/view?usp=drive_link

To verify model parameter integrity and test inference on any workstation, place `best_model.pth` into the `weights/` directory and run:

```
python verify_submission.py
```

An interactive forensic dashboard is also provided via `python app.py`.